Conditional Predicates – Part I: Syntax*

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Abstract

This paper develops an account of the structure of 'if'-clauses in noun phrases (*N if S*) and verb phrases (*V if S*), labeled 'conditional predicates'. The analyses are developed in a three-layered architecture for noun phrases (√-n-n*) and verb phrases (√-v-v*). I provide evidence from ellipsis and interactions with modifiers which support analogous structures for *N if S* conditionals and *V if S* conditionals. In both cases the 'if'-clause combines with a predicate and is interpreted under the base position of an external argument. I argue that interpretation at the level of the √P and nP/vP are each possible; and the 'if'-clauses can modify predicates of events, states, and entities. An array of novel corpus data and crosslinguistic data are provided. The data and syntactic results have rich implications for broader work on the semantics of conditionals as well as root meaning, ellipsis, and nominal and verbal predication.

1 Introduction

Hypothetical conditionals, it is traditionally said, are sentences "usually represented in English by the words 'if… then,' taking ordered pairs of propositions into propositions" (Stalnaker 1968: 98). Their "canonical form … is a two-part sentence," as in (1), consisting of an antecedent, 'school is cancelled', marked with 'if' and a consequent, 'the students will party', optionally marked with 'then' (von Fintel 2011: 1516).

(1) If school is cancelled, (then) the students will party.

Yet conditionals can also be formed with 'if'-clauses that follow a verb (*V if S*), as in (2), or a noun (*N if S*), as in (3) (Iatridou 1991, Lasersohn 1996, Bhatt & Pancheva 2017).

*"Part II: Semantics" in preparation. Email for current draft.

(2) The students will party if school is cancelled.

(3) The parties if school is cancelled will be crazy.

Though conditionals have been the source of extensive multidisciplinary research across subfields of linguistics, philosophy, computer science, and psychology, comparatively little attention has been given to sentences such as (2)–(3) with conditional predicates. Truth-conditions for sentences with *V if S* conditionals are generally left underived (e.g., Higginbotham 1986, von Fintel 1998, Leslie 2009, Klinedinst 2011). Treatments of *N if S* conditionals are embryonic (Lasersohn 1996: 160–163, Frana 2017: 64–69, Silk 2021: 178–180). This can lead to hasty conclusions and theories that are ill-equipped to capture a broader range of cases.

This paper works to address this state of affairs. The overarching aim is a more systematic investigation into the structure, meaning, and use of conditionals in noun phrases and verb phrases. Basic questions include:

- Should the Logical Forms of sentences with conditional predicates be modeled in the same way as the Logical Forms of *If… (then)* conditionals?
- Should the compositional semantics of sentences with conditional predicates be modeled in the same way as the compositional semantics of *If… (then)* conditionals?

I focus here exclusively on matters of syntax — specifically, on the structure of conditional predicates at LF, the level that is semantically interpreted. Critical issues include:

- Are there 'if'-clauses that combine with a verb phrase or noun phrase at LF?
- If so, where do 'if'-clauses combine in the verb phrase or noun phrase?

Examining these questions advances longstanding work on comparisons among nominal, verbal, and sentential domains. The present discussion may provide a basis for future work on the compositional semantics.

An overview of the paper is as follows. Building on previous work, §2 provides evidence that some 'if'-clauses are interpreted inside the verb phrase (§2.1) or noun phrase (§2.2). I develop the analyses in a layered architecture for verb phrases and noun phrases. The literature distinguishes, at minimum, phrasal projections of the verb/noun V/N and an element v*/n* that closes off the verb/noun phrase and introduces the external argument (if any), such as an agent/possessor. For *V if S* conditionals, I provide evidence from adverbial modification and verb phrase ellipsis which indicate that the 'if'-clause can combine under v* with VP. I provide evidence from restrictive relative clauses, nominal ellipsis, and quantified and possessive noun phrases which support an analogous structure for *N if S* conditionals. In both cases the 'if'-clause combines with a VP/NP predicate and is interpreted under the base position of an external argument.

§3 refines the preliminary analyses from §2. Despite the parallels between *V if S* and *N if S* conditionals, they cannot in general be reduced one to the other (§3.1) or derived from a common core (§§3.2–3.3). The architecture from §2 is updated to incorporate the distinction between the lexical root √ and an element v/n that marks the verbal/nominal environment. I argue that 'if'-clauses can combine both at the level of the √P and at the level of the vP or nP predicate (§§3.2–3.4). Data are provided from adverbial modification and event structure, and from v-stranding and n-stranding ellipsis crosslinguistically. §4 takes stock.

The main goal of this paper is an analysis of 'if'-clauses in nominal and verbal contexts. A subsidiary goal is to illustrate how insights and methods from diverse areas can be fruitfully integrated in theorizing. The proposed accounts build on developments in syntax and morphology on layered analyses of verb phrases and noun phrases (§§2–3). Previous work has focused primarily on the verbal domain. The extensions to noun phrases are comparatively underdeveloped. An array of novel crosslinguistic data are also provided from Persian, Finnish, and varieties of English and Japanese, which may be of broader interest. Though the kind of verbal ellipsis ("v-stranding" ellipsis) examined in §3 has begun to receive attention in wider literature, there remains nearly no investigation into the phenomenon in noun phrases ("n-stranding" ellipsis). The relevant dialects of Japanese have been largely unexamined. While there has been little work on *N if S* conditionals since Lasersohn (1996) introduced them into the literature, we will see that *N if S* conditionals are well attested in English corpora and crosslinguistically (§§2.2.5, 3.1, 3.3). The data provided should, I hope, be fruitful for future syntactic and semantic work, independent of the particular accounts developed here.

A preliminary remark: It is important to read examples with a smooth non-parenthetical intonation contour, without a pause before the 'if'-clause, unless explicitly indicated otherwise. Lasersohn (1996: 155–156) observes that 'if'-clauses can appear in certain sentence-internal positions only when set off by parenthetical intonation, as in (4)–(5) with an 'if'-clause following a bare proper name or auxiliary verb. The (b)-examples are anomalous without parenthetical intonation, as marked by a comma in the (a)-examples.

(4) a. John, if you bother him long enough, will give you five dollars.
    b. *John if you bother him long enough will give you five dollars.

(5) a. John will, if you bother him long enough, give you five dollars.

b. *John will if you bother him long enough give you five dollars.

(Lasersohn 1996: exs. 4–5)

No such displacement, Lasersohn observes, is necessary in (6), or, we can add, (2)–(3), where the 'if'-clause is immediately adjacent to a noun phrase or verb phrase.

(6) The <u>price if you pay now</u> is predictable; the price if you wait a year is not.

(adapted from Lasersohn 1996: ex. 6b)

Reading the 'if'-clause parenthetically, as if set of by a comma, marks a structural difference (see also Dehé 2014: ch. 2). Only (7a) can receive the relevance-conditional reading which implies that Timmy looks tired. The sentence-final 'if'-clause in (7a) is not part of sentence's the main predicate (e.g., Iatridou 1991: 56–57).

(7) a. Timmy looks tired, if you want my opinion.

b. #Timmy looks tired if you want my opinion.

I put aside conditionals in which the 'if'-clause is extraposed and set off by parenthetical intonation.

2 Conditional predicates as predicates

This section develops a preliminary account of 'if'-clauses in verb phrases and noun phrases as predicate modifiers. In the (a)-examples in (8)–(9) we see the 'if'-clause follow a complex verb phrase and noun phrase. The 'if'-clause cannot intervene between the relational verb or noun and its internal argument (cf. (4b)–(5b)).

(8) a. Alice will high-five Bert if school is cancelled.

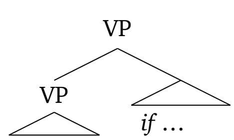
b. *Alice will high-five if school is cancelled Bert.

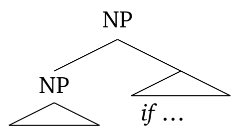
(9) a. the result of the game if we don't call time-out

b. *the result if we don't call time-out of the game

A natural hypothesis is that the 'if'-clause combines at the level of the phrase, as schematized in (10).

(10) a.



b. 

The following sections examine syntactic tests and crosslinguistic data to sharpen this idea.

2.1 'If'-clauses in verb phrases

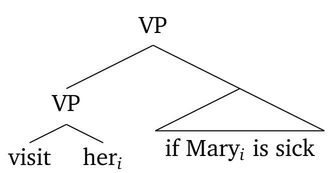
Let's start with *V if S* conditionals. I begin with previous arguments that sentence-final 'if'-clauses may attach to the VP (see Iatridou 1991: 9–33, Bhatt & Pancheva 2017: 9–13).

2.1.1 Condition C

First, Iatridou (1991: 10–11) considers binding data such as (11) to test the position of the 'if'-clause relative to the main clause. Iatridou observes that the coreferential reading, where the pronoun ('she'/'her') refers to Mary, is unavailable in (11a) yet available in (11b).

(11) a. *She$_i$ yells at Bill if Mary$_i$ is hungry.
b. Bill visits her$_i$ if Mary$_i$ is sick. (Iatridou 1991: exs. 2a, 4a)

This contrast is expected if the 'if'-clause is c-commanded by the subject of the main clause though not by the direct object. The anomalousness of (11a) is diagnosed as a violation of Condition C, which excludes pronouns from coreferring with non-pronouns which they c-command (e.g. *'She$_i$ knows that Alice$_i$ won'; Chomsky 1981). The acceptability of (11b) is expected if the 'if'-clause combines with the VP, per (10a). The direct object in (12) doesn't c-command the proper name in the 'if'-clause.

(12) 

What is at issue for our purposes, recall, is the structure which is semantically interpreted (§1). So, for (11) to be probative, Condition C must apply at LF. Examples such as (13) with an unaccusative ('fall') pose a challenge.

(13) *She$_i$ falls if Mary$_i$ is dizzy.

It is generally accepted that the derived subject in an unaccusative structure originates in the VP, in the same position as the direct object in a transitive structure (Perlmutter 1983, Burzio 1986, Marantz 2013, Harley 2014; cf. Borer 2005). If non-semantically-driven movement is undone at LF, 'she' in (13) is interpreted in the same position as 'her' in (11b); each is the internal argument of the verb:

(14) a. [$_{\text{VP}}$ visit her$_i$]
b. [$_{\text{VP}}$ fall she$_i$]

If Condition C applies at LF, one might expect judgments for (11b) and (13) to be on a par. The pronouns are in the same position, thus in the same position relative to the 'if'-clause, wherever it is. Yet (11b) is acceptable, (13) degraded (again ensuring a consistent, non-parenthetical intonation). The examples in (15) with the alternating verb 'shake' illustrate the contrast vividly.

(15) a. *She$_i$ shakes if Mary$_i$ is cold.
b. *She$_i$ shakes Bill if Mary$_i$ is cold.
c. Bill shakes her$_i$ if Mary$_i$ starts snoring.

The derived subject in the unaccusative configuration ((13), (15a)) patterns with the external subject in the transitive configuration ((11a), (15b)), not with the direct object in the transitive configuration ((11b), (15c)). This is unexpected if Condition C applies at LF.

The Condition C data from Iatridou provide evidence that, at some level of representation, the sentence-final 'if'-clause is c-commanded by a main-clause agentive subject and is not c-commanded by a main-clause direct object. The examples with unaccusatives give reason to question whether this level is LF, given several plausible assumptions: (i) that Condition C constrains c-command relations; (ii) that the derived subject in an unaccusative structure originates in the same position as the direct object in a transitive structure (or at least originates in some other position that doesn't c-command an adjunct of VP); and (iii) that items that move for non-semantic reasons are semantically interpreted in their base position. Such assumptions aren't unassailable. For instance, one might question (iii) by appealing to arguments against reconstruction of A-movement (Chomsky 1995, Lasnik 1999; see von Fintel & Iatridou 2003, Sportiche 2017 for discussion). Yet for present purposes let's consider other evidence.

2.1.2 *VP constituency*

V if S conditionals can be targeted by phenomena known to target the verb phrase (cf. Iatridou 1991: 11–12, Bhatt & Pancheva 2017: 9–10). In (16) the conditionalized 'party if school is cancelled' is targeted by verb phrase ellipsis (marked by strikethrough). In (17) it provides the antecedent of the anaphor 'so'. In (18) it is topicalized by verb phrase fronting.[1]

(16) The students will party if school is cancelled, and the teachers will ~~[$_{VP}$ party if school is cancelled]~~, too.

(17) The students will party if school is cancelled, and …

a. the teachers will do so too.

b. so will the teachers.

(18) I said that the students will party if school is cancelled, and party if school is cancelled they will.

Such tests provide "[c]lear evidence that sentence-final *if*-clauses are constituents of the VP" (Bhatt & Pancheva 2017: 9).

Like in §2.1.1, one cannot simply read off an LF from a constituency test. The fact that 'Bert' is not elided in (19), with unaccusative 'die', doesn't show that the derived subject isn't semantically interpreted in the VP.

(19) Alice will die if there is school tomorrow, and Bert will ~~[$_{VP}$ die if there is school tomorrow]~~, too.

Yet there are well studied accounts of how the derived subject comes to be extracted from the ellipsis site and pronounced. On one view, 'Bert' moves for Case reasons before the VP is marked to be elided at PF (cf. Aelbrecht 2010, Merchant 2013). The non-semantically-driven movement is undone at LF, and 'Bert' is semantically interpreted in its base position as the internal argument of 'die' (cf. (14b)).

So, examples such as (16)–(19) provide evidence that some sentence-final 'if'-clauses are part of the verb phrase. A conservative conclusion is that the conditionalized predicate is also a constituent at LF and semantically interpreted.

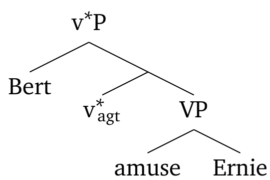
2.1.3 *External arguments*

The previous discussion agrees with previous syntactic work that some sentence-final 'if'-clauses attach within the sentence's main predicate (see also Haegeman

[1]Verbal fronting is generally marked in English. What is important is the example's relative acceptability.

2003). It will be important to delineate the structure of the conditionalized predicate more precisely. It is common in light of substantial morphosyntactic and semantic evidence to treat what we have been calling “VP” as a sequence of layered projections. Notable for present purposes is the distinction between the verb V and an element v* (or Voice) that closes off the verb phrase and, in agentive sentences, introduces the external argument (terminology varies).[2] A structure for transitive ‘Bert amuse Ernie’ is in (20). (I use ‘v^*_{agt}’ for the unpronounced external-argument-selecting v* head).

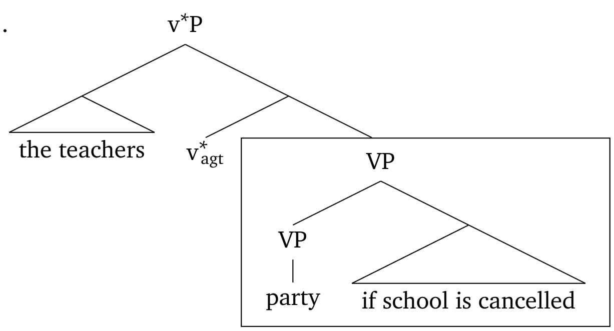
(20)



The agentive subject originates outside the VP.

There are reasons to think that sentence-final ‘if’-clauses combine with the VP, in the present narrower sense. First, it has been argued that the target of verb phrase ellipsis is under, i.e. c-commanded by, v* (in my notation; Johnson 2008, Aelbrecht 2010, Merchant 2013, though see Baltin 2012). If this view is correct, then a straightforward interpretation of ellipsis data such as (16) is that the ‘if’-clause is also under v*. The box around the complement of v* in (21) indicates the ellipsis site, which includes the ‘if’-clause.

(21) … and …



Second, ‘if’-clauses can be followed by modifiers that have been argued to adjoin under v*. Examples with a durative ‘for’ adverbial and VP adverb are in (22)–(23)

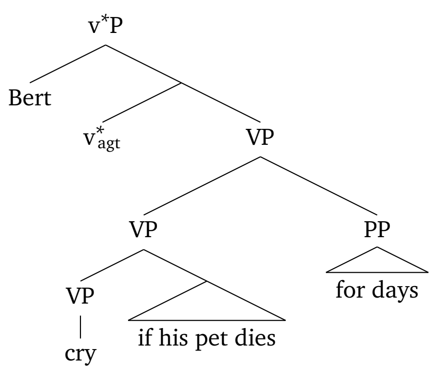
[2]For a useful overview, see Harley 2017; see also Chomsky 1995, Kratzer 1996, Marantz 1997, Borer 2005, Pylkkänen 2008, Baltin 2012, Harley 2013, Wood & Marantz 2017, Silk 2021. In a phase-based syntax (Chomsky 2001, 2008), v*, in my terminology, is the head demarcating the edge of the verbal phase. We will consider further subdivisions in §3.1.

(cf. Sportiche 1988, Ernst 2002: chs. 5–6). The conditionalized predicate ‘cry if his pet dies’ in (22b) is further modified by the aspectual predicate (‘for days’). The adverb in (23a) characterizes the manner of the ringing-of-the-bell event if there’s a fire.

(22) a. Alice will play video games if school is cancelled for hours.
 b. Bert will cry if his pet dies for days.

(23) a. I assure you: Alice will ring the bell if there’s a fire very loudly.
 b. The sergeant will notify you if there is a threat nonverbally.

The structure for (22b) in (24), where the ‘if’-clause adjoins to the VP, captures the linear order.

(24)


The previous data with adverbials in English are corroborated by crosslinguistic data. For instance, Karimi (2003, 2005) has argued that there is a class of low adverbs in Persian which mark the left edge of the verb phrase — in the present terminology, the v*P. Karimi observes that specific subjects can follow such adverbs, as in (25) with *kâmelan* ‘completely’ (see 2005: 124–126 for additional examples; Persian is verb-final with a basic SOV word order). Unlike in languages such as English, the subject can be pronounced inside v*P.

(25) kâmelan [$_{v^*P}$ Kimea be hush umade].
 completely Kimea to sense come-3SG
 ‘Kimea has completely regained her senses.’
 (adapted from Karimi 2005: 94; Persian)

Karimi's observation extends to agentive subjects. The subject in (26) remains in its base position.[3]

(26) hargez [$_{\text{v*P}}$ bache-hâ dâd ne-mizanan].
never child-PL yell NEG-hit.PRES.3PL
'Children never yell.' (Persian)

Notably, 'if'-clauses can follow the v*P-edge adverb and subject:

(27) hargez [$_{\text{v*P}}$ bache-hâ [agar khoshhâl bashan] dâd ne-mizanan].
never child-PL if happy be.SUBJ.3PL yell NEG-hit.PRES.3PL
'Children never yell if they are happy.'

(28) hamishe/hanuz [$_{\text{v*P}}$ Rostam [agar sir dashte bashim] ashpazi mikone].
always/still Rostam if garlic have.PART be.SUBJ.1PL cooking do.PRES.3SG
'Rostam always/still cooks if we have garlic.' (Persian)

The 'if'-clause's position inside v*P is indicated by the v*P-edge adverb. A conservative analysis is that the 'if'-clause is interpreted in this position under the subject.

A key takeaway from this section is that 'if'-clauses can originate in a position under (i.e. c-commanded by) the base position of an external argument. This will be important for work on the semantics.

2.2 'If'-clauses in noun phrases

§2.1 provided evidence that 'if'-clauses in *V if S* conditionals can be interpreted inside a verb phrase. Let's turn now to *N if S* conditionals such as (3).

(3) The parties if school is cancelled will be crazy.

[3] Thanks to Mohsen Moghri for assistance with the data.

N if S conditionals have received comparatively little attention.[4] Yet they are not a marginal phenomenon. We will see that *N if S* conditionals are well attested in corpora as well as crosslinguistically.[5]

2.2.1 DP contexts

First, *N if S* conditionals can occur in contexts where the 'if'-clause is evidently interpreted inside a DP. For instance, Lasersohn (1996: 154, 156) notes examples where the 'if'-clause occurs in a DP complement of 'know'. Examples with several other predicates that subcategorize for DPs are in (29). A naturally occurring example with 'imagine' is in (30).

(29) a. The students are discussing/anticipating/excited about the parties if school is cancelled.
  b. We await the parties if school is cancelled with eager expectation.

(30) Imagine the riots if they overturn the cases that uphold gay marriage rights.[6]

The discussions, etc. described in (29) and the exhortation expressed in (30) are not conditional. The 'if'-clauses are interpreted inside the definite descriptions.

Frana (2017: 51) oberves that *N if S* conditionals can occur in a syntactic island — an environment out of which material generally cannot move. The adjunct in (31) and presuppositional DPs in (32) are islands for extraction.

(31) a. Given the traffic if there is an accident, we should leave early.
  b. *What given should we leave early?

[4]The studies following Lasersohn's (1996) SALT paper that I am aware of are Shizawa 2011: ch. 6 (which examines usage of *N if S* conditionals at an informal level), Frana 2017 (which considers how an NP-modifier analysis may be integrated in a Kratzer-style restrictor semantics), Blümel 2019 (which provides evidence for an NP-modifier syntax in German), and Silk 2021: §8.3 (which focuses on the compositional semantics in a framework with variables for assignment functions). Declerck & Reed's (2001) "comprehensive empirical analysis" of conditionals includes one set of examples (p. 369). Bhatt & Pancheva (2017) mention *N if S* conditionals but restrict their attention to reviewing several of Lasersohn's examples.

[5]Dixon (1972) reports the adnominal structure to be the primary means of expressing hypothetical conditionals in languages such as Dyirbal:

(i) bayi yaɽa ɽudu balga-ŋu guyibi-ñ
 CL1.M man hollow hit-REL die-FUT
 'If a man is hit in the hollow in the back of his neck, he will die.' (lit. 'A man if hit … will die')
 (from Dixon 1972: 362; Dyirbal (Pama–Nyungan))

[6]www.reddit.com/r/teenagersnew/comments/vjvzs4/imagine_the_riots_if_they_overturn_the_cases_that (retrieved from the Timestamped JSI Web Corpus via Sketch Engine)

(32) a. Alice said you recorded most/both parodies of the riots if Oscar wins the election.

b. ?*What did Alice say you recorded most/both parodies of?

In (31) the ‘if’-clause cannot move out of the ‘given…’ adjunct to scope over the main clause. It must be interpreted in situ.

The naturally occurring example in (33) combines these features The conjunction of definite descriptions provides the complement of ‘weigh’. A similar example from Lasersohn is in (34). Here, “neither *if*-clause takes scope over the rest of the sentence, but only over its own conjunct noun phrase” (Lasersohn 1996: 156).

(33) If you want to know if school staff should be armed, then you might want to learn from the cafeteria workers, custodians, teachers, principals, and school superintendents who volunteered to put their body between our kids and a bullet. They weighed the costs if they were armed and the costs if they were unarmed. They make the decision to be an armed protector every day.[7]

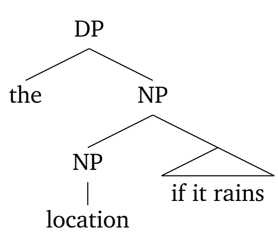
(34) The location if it rains and the location if it doesn’t rain are within five miles of each other. (Lasersohn 1996: ex. 10)

Lasersohn (1996: 156) concludes, “There seems no alternative in this construction to regarding the *if*-clauses as forming part of the noun phrases.”

The previous data provide evidence that ‘if’-clauses can be interpreted inside a noun phrase, informally understood. However, the data leave open whether the ‘if’-clause combines with a predicate. Both structures in (36) for the first conjunct of (35) are possible.

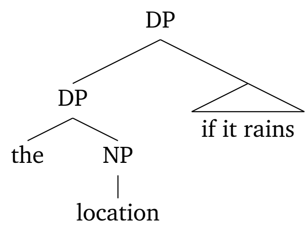
(35) [[$_{DP}$ The location if it rains] [and [$_{DP}$ the location if it doesn’t rain]]] …

(36) a.



[7] www.ammoland.com/2018/08/who-says-we-should-arm-school-staff-to-protect-our-children/ (retrieved from the Timestamped JSI Web Corpus via Sketch Engine)

b.



In (36a) the ‘if’-clause combines with an NP to form a complex predicate. In (36b) it combines with a DP.

2.2.2 *NP modification*

There is one example of Lasersohn’s, in (37), which supports the structure from (36a) where the ‘if’-clause combines with a nominal predicate. §2.1.3 used adverbials to help locate ‘if’-clauses in *V if S* conditionals. Modifiers such as restrictive relative clauses can play an analogous role with *N if S* conditionals. The restrictive relative clauses in (37)–(38) follow the ‘if’-clause.

(37) The consequences if we fail that he mentioned are not nearly as bad as the consequences if we fail that he didn’t mention. (Lasersohn 1996: ex. 11)

(38) There will be many riots if Oscar wins. But there will be few riots if Oscar wins that cause serious damage.

Consider Lasersohn’s (37). There will be various consequences if we fail. Some were mentioned; some were not. To yield the intended interpretation, ‘the’ must combine with the *N if S* conditional as restricted by the relative clause. The requisite structure for the definite description is a structure like (36a), where the ‘if’-clause combines with the head noun ‘consequences’, as reflected in (39).

(39) The [[$_{\text{NP}}$ consequences if we fail] that he mentioned] …

Restrictive relative clauses do not combine with DPs (cf. Partee 1975: 231, Demirdache 1991: ch. 3, Bhatt 2002: 70–71).

2.2.3 *NP constituency*

In §2.1.2 we saw how *V if S* conditionals can be targeted by phenomena such as verb phrase ellipsis and anaphora. Parallel phenomena can be observed with *N if S* conditionals. In (40) the conditionalized ‘party if school is cancelled’ is targeted by noun phrase ellipsis. In (41) the *N if S* conditional is targeted both by anaphoric ‘one’ replacement and by restrictive modification (§2.2.2).

(40) a. Alice's party if school is cancelled will be better than Bert's ~~$[_{\text{NP}}$ party if school is cancelled$]$~~.

b. Most parties if school is cancelled will be fun, but some ~~$[_{\text{NP}}$ parties if school is cancelled$]$~~ will be lame.

(41) The party if school is cancelled that Alice told me about will be better than the one that Bert told you about.

Like in §2.1.2, a conservative conclusion is that the conditionalized predicate is also a constituent at LF.

One might wonder whether the 'if'-clause needs to be included in the ellipsis site and antecedent in (40)–(41) to capture the intended interpretations. Suppose the interpretation of presuppositional noun phrases is contextually restricted by a domain variable D (cf. von Fintel 1994, Stanley & Szabó 2000). For (40b), suppose that only 'party' is targeted by ellipsis and the 'if'-clause contextually constrains the interpretation by providing a value for the domain variable, as reflected informally in (42), letting '$\cdots$' represent the contextual domain restriction. (To fix ideas I assume a single variable attached to the determiner.)

(42) a. $[_{\text{DP}}$ some$_D$ ~~$[_{\text{NP}}$ parties$]$~~$]$ will be lame

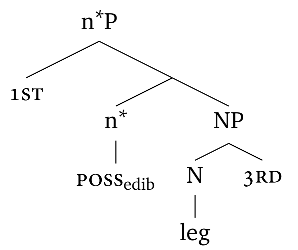
b. $\exists x\colon \text{party}(x, w) \wedge \cdots \wedge \text{lame}(x, w)$

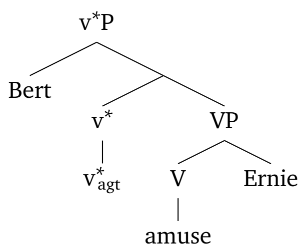
Such an analysis can be put aside. The 'if'-clause in (40b) is not a restrictive modifier. (40b) needn't imply that there are or will be parties.

2.2.4 External arguments

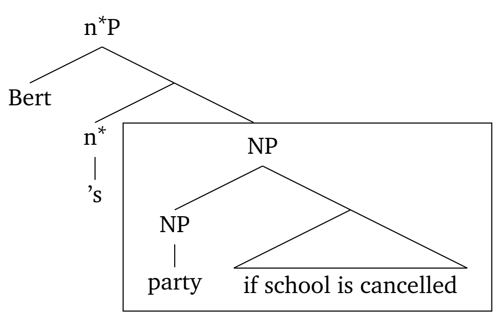
The previous sections provided evidence supporting a structure for *N if S* conditionals that parallels the structure for *V if S* conditionals, where the 'if'-clause forms a complex predicate with the noun phrase or verb phrase. §2.1.3 refined the analysis of *V if S* conditionals in a layered architecture for verb phrases. Parallels in the nominal domain have been used to support layered analyses of noun phrases (Hiraiwa 2005, Chomsky 2007, Lecarme 2008, Megerdoomian 2008, Wood & Marantz 2017, Silk 2021; notation varies). Notable again is the distinction between the (here) noun N and an element n* that closes off the noun phrase and introduces an external argument, if present. In English, n* can be realized by the genitive morpheme *'s*, which may introduce a thematic "possessor" argument. The structure in (44) for the complex possessive in (43) parallels the structure for active transitives in (20), reproduced in (45) (see Silk 2021: chs. 6–7 for extended discussion).

(43) nək-k nelk-n
POSS$_{\text{edib}}$-1SG leg-3SG.GEN
'my leg of its (e.g., of chicken, to eat)'
(Lynch 1973: 91; Lenakel (Austronesian))

(44)


(45)


Suppose that noun phrase ellipsis in English targets a phrase under n* (cf. Saab 2019). The ellipsis in (40a) then includes the 'if'-clause, as reflected in (46).[8] The 'if'-clause combines with NP under the thematic possessor ('Bert'), analogously to how the 'if'-clause in (21) combines with VP under the agentive subject. (The box again indicates the ellipsis site.)

(46)


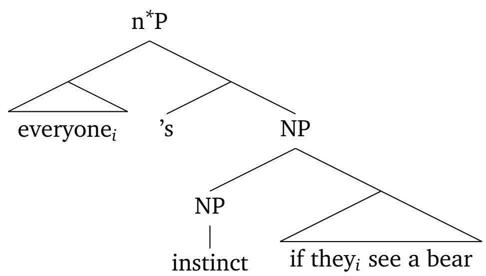
This analysis predicts that quantified possessors should be able to bind pronouns in an *N if S* conditional. This prediction is borne out, as seen in (47). The quantifier phrase in (48) c-commands the 'if'-clause.

[8]I leave open how the pronounced forms of English prenominal genitives are derived in the narrow syntax and morphophonolological components.

(47) a. Everyone$_i$'s instinct/reaction if they$_i$ see a bear is to run away.

b. Every student$_i$'s first thought if they$_i$ get a bad grade is that the teacher made a mistake.

(48)



Further evidence that the ellipsis site is under n* comes from (what we might call) "possession mismatches" between an elided noun phrase and its antecedent.[9] In possessives the genitive morpheme *'s* introduces a thematic possessor as an external argument; however, not all n* heads introduce a possessor argument. A first example is n*Ps in quantifier phrases. Though n* is unpronounced in English quantifier phrases, its effects can be observed in other languages. n* can be identified with the functional projection posited for assigning genitive Case in quantifier complements in certain Slavic languages, as in (49) (Bošković 2006, 2014; see also Silk 2021).

(49)

Mnogo	ljudi	želi	pravdu.
many	people.GEN	want	justice

'Many people want justice.'

(from Hartmann & Milićević 2009: ex. 14; Serbian)

The Case-assigning n* head is implicit and doesn't introduce a thematic possessor.

A second example is n*Ps in non-possessive readings of genitives. In a possessive such as 'Alice's party', 'Alice' is interpreted as the subject of a broadly possessional relation, as reflected in (50). In (51), by, contrast, 'Alice' is interpreted as the object of the mother relation expressed by the head noun (e.g., Partee 1983, Barker 1995: ch. 2, Partee & Borschev 2003, Silk 2021: ch. 6). Although English uses the same item *'s* for both readings, the distinction is explicit in many languages (cf. (43)).

(50) Alice's party (≈ "the o such that o is a party and Alice bears R_{poss} to o")

(51) Alice's mother (≈ "the mother of Alice")

[9] We will look at additional crosslinguistic evidence in §3.3.

In genitives such as (51), the genitive morpheme realizing n* doesn't assign a thematic possessor role.

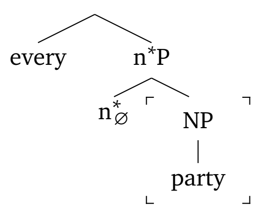
"Possession" mismatches with such n* heads are possible in noun phrase ellipsis. Consider, first, mismatches involving possessives and quantifier phrases, as in (52)–(53). In (52) the noun phrase containing the ellipsis is possessive, and the noun phrase containing the antecedent is non-possessive; vice versa in (53).

(52) I didn't go to every party, but I went to Alice's.

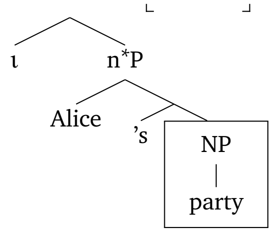
(53) Alice's party was better than most.

Treating the ellipsis site as being under n* provides an explanation. The n* head isn't part of the material that needs to be relevantly identical in the elided structure and its antecedent in order for ellipsis to be licensed, as reflected in (54) for (52).[10] (I use the open box to indicate the antecedent; the solid box again indicates the ellipsis site. I use '$n^*_{\varnothing}$' for the implicit head in quantifier phrases. I assume, following Partee 1983, that English prenominal genitives *X's Y* include an implicit definite determiner, here written 'ι'.)

(54) a. I didn't go to



b. but I went to



[10]The Case-assigning properties of n* observed in (49) continue to be manifested in ellipsis, as in (i). (Thanks to Petar Milin for assistance with the data. The examples come from the Serbian region, though the phenomenon occurs across Bosnian-Croatian-Serbian.)

(i) a. Mnogo penzionisanih ljudi je glasalo, ali mnogo mladih nije.
many retired.M.GEN.PL people.GEN AUX voted but many young.M.GEN.PL NEG.AUX
lit. 'Many retired people voted, but many young didn't.'

b. Mnogo malih pasa je lajalo i nekoliko velikih takođe.
many small.M.GEN.PL dogs.GEN AUX barked and a few big.M.GEN.PL too
lit. 'Many small dogs barked, and a few big too.' (Serbian)

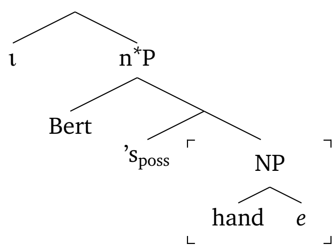
Since the target of the ellipsis doesn't include n*, the heads *'s* and $n^*_\varnothing$ can differ without violating the identity condition on ellipsis.

Possession mismatches involving possessive vs. non-possessive readings of genitives are possible as well, as in (55). The relevant interpretation in the context is that the hand of the cadaver that Bert is practicing on is broken worse than Alice's own hand.[11] The posited structures for the noun phrases are in (56). (I assume that the genitive subject in non-possessive readings is derived and originates as an internal argument of a relational head noun. I assume that the null indefinite argument in 'Bert's hand' is syntactically present (cf. (43)), here represented as *e*. The subscripts 'poss' and 'nposs' are used to mark the distinction between possessive and non-possessive *'s*, respectively.)

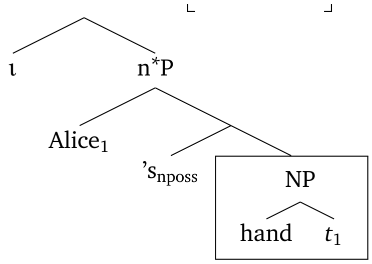
(55) [Context: We're medical students. We're talking about the different cadavers people are practicing on. Alice recently fractured her hand.]

Bert's hand is broken even worse than Alice's after the skiing accident.

(56) a.



b.



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[11]Such mismatches can also be observed in languages where the difference between possessive and non-possessive readings is marked overtly, as in (i). The noun phrase containing the antecedent uses a genitive construction, and the noun phrase containing the ellipsis uses the possessive. (Thanks to Vladimir Borschev, Sergey Kruk, and Natasha Rulyova for assistance with the data.)

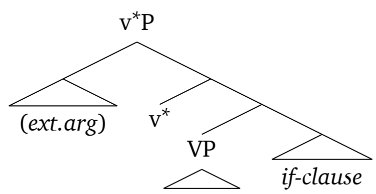
(i) [Context: Petya has a murderer in his crew that he hires to do his dirty work for him. Masha was recently murdered by someone else, a stranger.]

Ubiitsa Mashi opasnee, chem Petin ~~ubiitsa~~.
murderer Masha.GEN more.dangerous than Petya.POSS murderer
'Masha's murderer is more dangerous than Petya's.' (Russian)

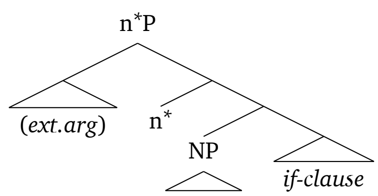
The analyses in (56) satisfy the identity condition on ellipsis assuming that traces can be relevantly equivalent to unexpressed arguments for purposes of ellipsis licensing (Merchant 2013).[12] The identity condition would be violated, however, if the ellipsis site was higher than NP given the different features on n*.

The previous data support the analysis of elided *N if S* conditionals in (46) in which the 'if'-clause originates under n* and the position of a thematic possessor, determiner, or quantifier. A parallel structure emerges for *V if S* conditionals (§2.1.3) and *N if S* conditionals:

(57) a.



b.



In each case, the 'if'-clause combines with a VP/NP predicate and is interpreted in a position c-commanded by an external argument. Both points have significant implications for the semantics.

2.2.5 *Licensing?*

There has been little work to date on the syntax and semantics of *N if S* conditionals (n. 4). Perhaps due to the paucity of examples, several authors have posited licensing conditions on uses of 'if'-clauses in noun phrases — e.g., that *N if S* conditionals

[12] Considering examples of voice mismatches in verb phrase ellipsis, Merchant (2013: 91) writes that "the trace of an element moved out of an elided phrase … need not be featurally identical" to its correlate in the antecedent "if the differing featural content can be recovered by an element outside the ellipsis site." In (56), the correlate in (56a) of the trace *t* is the null argument *e*, and the different content is given by 'Alice', moved out of the ellipsis site in (56b).

can only occur with certain types of nouns or in certain environments.[13] Corpus data suggests that there are no such constraints.

First, many of the *N if S* conditionals thus far have used nouns that describe events (e.g., 'party'). Yet 'if'-clauses can modify items in various semantic classes. *N if S* conditionals are possible with nouns characterizing events ((58)), states ((59)), and other entities ((60)). Some of the event- or state-denoting nominals have verbal or adjectival cognates; many do not. (I use a simple 'Imagine DP' frame to focus attention on the conditionals.)

(58) *Event-denoting*

a. Imagine the riots/fights/parties if the Rangers win. (They will be crazy.)
b. Imagine the acceptance speech if [the Oscar for makeup and hair] went to "Jackass Presents: Bad Grandpa."[14]
c. Imagine the kerfuffle if I had piggybacked him through the Long Room![15]
d. Imagine the catastrophe if this country should have a fast-moving infectious threat such as smallpox.[16]
e. Imagine the deaths if the shooter had a silencer.[17]

(59) *State-denoting*

[13] Shizawa (2011) offers the following:

> *The Licensing Condition of Adnominal Conditionals*
> The referent of a noun modified by the *if*-clause must be one that can be construed either semantically or pragmatically as having a resultant value to be determined by the fulfillment of the condition in the *if*-clause. (Shizawa 2011: 156–157)

The talk of a referent's "resultant value" isn't made formally precise. The idea seems to be an instance of the general conversational norm that modifiers should nontrivially modify the phrase they are modifying. Frana (2017: 51, 56) notes Lasersohn's (1996: 161) claim that *N if S* conditionals are generally anomalous with proper names qua rigid designators. The idea is again the general point that modifiers (here, 'if'-clauses) should play a nontrivial semantic role. Blümel (2019: 1) claims that "only intensional nouns co-occur with, e.g. can be modified by, adnominal conditionals." It isn't said what is meant by 'intensional noun'.

[14] www.latimes.com/entertainment/movies/moviesnow/la-et-mn-sharkey-oscar-notebook-story.html (modified from 'Oh, but imagine…')

[15] www.thehindu.com/opinion/columns/richard-dawkins-on-collegemate-tiger-pataudi-sublime/article33504006.ece

[16] "...The evident lack of preparation and coordinated response is appalling and frightening" (2014). www.washingtonpost.com/opinions/missing-key-points-on-ebola/2014/10/17/e8066108-53e6-11e4-b86d-184ac281388d_story.html

[17] https://twitter.com/HillaryClinton/status/914853465926639618

a. Imagine our love for each other if we get married. (It will last forever.)
b. Imagine his mood later if he doesn't nap. (It will be terrible.)
c. Imagine the crisis if the factory shuts down.[18]
d. Imagine a stamp collector's joy if CNN launched into breathless coverage of a philatelic convention.[19]
e. Imagine the national outrage if 50,000 koalas were dropping dead from the trees.[20]

(60) *Entity-denoting*
a. Imagine our country if Trump is re-elected. (It may not survive.)
b. Imagine the costs if we have to build a seawall around Manhattan, just to protect against storms and rising tides.[21]
c. Imagine his face if she told him the truth.[22]
d. Imagine the banner headlines if a marine biologist were to discover a species of dolphin that wove large, intricately meshed fishing nets, twenty dolphin-lengths in diameter![23]
e. Imagine the hype videos if Snoop Dogg was on board.[24]

Second, *N if S conditionals* can occur in noun phrases that are definite, indefinite, or quantified, with weak or strong quantifiers, as in (61)–(62) (cf. (59d)). The restrictive relative clauses in the (b)-examples indicate that the 'if'-clauses are modifying the noun phrase, not 'be' or the verbal predicate (cf. §2.2.2).

(61) a. Most/All symptoms if you're still contagious are severe.
b. There are (∅/no/some/a few/many) symptoms if you're still contagious that are severe.

(62) a. Most/All board members if Oscar is elected will resign.

[18] https://news.band/border-closure-nigerian-rice-millers-have-closed-shop-dantata
[19] www.vox.com/first-person/2018/10/26/18019920/lottery-tickets-mega-millions-power-ball
[20] www.newsmaker.com.au/news/28832/mass-dieoffs-in-extreme-heat-spark-call-for-urgent-government-action-to-end-war-on-flyingfoxes (modified from 'You can imagine…')
[21] www.bbc.com/future/article/20170418-how-western-civilisation-could-collapse
[22] Retrieved from the British National Corpus
[23] R. Dawkins (2016), *The selfish gene*, Oxford University Press, 4th edn., p. 308 (modified from 'Just imagine…')
[24] www.dallasnews.com/sports/2014/01/14/snoop-dogg-s-son-a-4-star-wide-receiver-reportedly-interested-in-baylor (modified from 'Can anyone else imagine…?')

b. There will be (∅/no/some/a few/many) board members if Oscar is elected who will resign.

Third, many of our examples have used *N if S* conditionals in broadly modal (future-oriented, generic) contexts, as in (63).

(63) a. His cough if he is still contagious will be phlegmy. (So watch out later.)
b. His cough if he was still contagious would be phlegmy. (So he's probably not contagious.)
c. His cough if he is still contagious is probably phlegmy. (So why don't you see how he's doing?)
d. The cough if you are contagious is phlegmy. (That's how it was with Alice and Bert.)

This too is inessential. As Lasersohn (1996: 158) observes, "there are other examples in which it is clearer that an adnominal *if*-clause is positioned in an extensional context" (see also Frana 2017: 54–55). A realtor might utter (64) in attempting to seal a deal with a potential client. A speechwriter might utter (65) to a politician awaiting the results of an election.

(64) Your new home if you sign with us is right in front of you.

(65) Your/Every speech if you win is in the green folder. Your speech if you lose is in the red folder.

The subject-position *N if S* conditionals in (64)–(65) have existential import. (64) implies that something is right in front of you; (65) implies that something is in the green folder and something in the red folder. Yet the fact that a noun phrase occurs in an extensional context needn't imply that the sentence "make[s] a … claim about" something in "the actual world" (Frana 2017: 55). Elements inside the noun phrase may shift its denotation. Consider (66).

(66) Most returns if Alice was working were processed immediately. (So hopefully she was working earlier, and we won't have to deal with many complaints.)

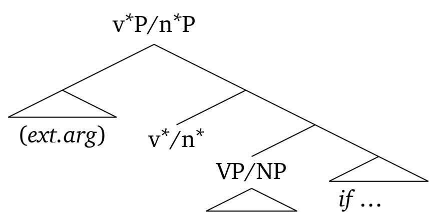
(66) doesn't imply that anything was processed immediately. Yet the perfective does not create an intensional context.

There do not appear to be interesting syntactic or semantic constraints on the kinds of nouns that can be modified by 'if'-clauses or the linguistic contexts in which *N if S* conditionals can appear. *N if S* conditionals can occur with nouns that characterize events (‘party’), states (‘mood’), and entities (‘home’); with nouns that are morphologically underived (‘riot’) and derived (‘reaction’); in noun phrases that are definite, indefinite, and quantified, with weak and strong quantifiers; and in intensional and extensional contexts. An account of distributional differences may proceed at the level of sentences’ specific structure and conventional meaning and general norms of language use.

3 Refining the analyses

§2 argued that some *V if S* conditionals and *N if S* conditionals are interpreted as conditionalized predicates. There are ‘if’-clauses that combine with a verb phrase or noun phrase at LF. In particular, we saw evidence that ‘if’-clauses can be interpreted under the head that introduces an external argument, such as a thematic agent or possessor, as reflected in the analyses in (57), reproduced in (67).

(67)



This section further considers the extent of the parallels by attending to more detailed structure in verb phrases and noun phrases.

3.1 Nominalization (and Verbalization)

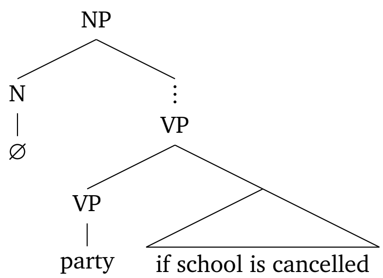
Our initial examples of conditional predicates in (2)–(3), adapted in (68)–(69), each used ‘party’ as the main verb or noun.

(68) Alice will <u>party if school is cancelled</u>.

(69) Alice’s <u>party if school is cancelled</u> will be crazy.

One might wonder whether the ‘if’-clause in (69) in fact combines with a noun phrase. How do we know the ‘if’-clause doesn’t modify an underlying verbal predicate which is then nominalized (cf. Kiparsky 1982), say along the lines in (70)?

(70)



A structure like (70) may be appropriate for some nominalizations. For instance, it has been argued that “ACC-*ing*” nominalizations embed verbal categories (Kratzer 1996, Harley 2009). The nominalization ‘PRO/them driving us around’ in (71) can be modified by an adverb (‘slowly’); and it licenses accusative case on the object (‘us’), characteristic of v* in active transitives.

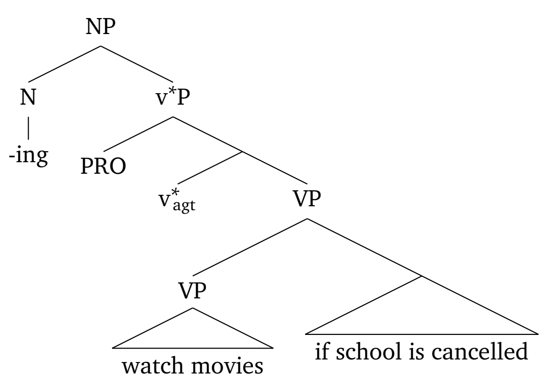
(71) I remember (them) slowly driving us around.

Like with the *V if S* conditionals from §2.1.3, the verbal modifiers in (72)–(73) can follow the ‘if’-clause. This provides evidence that the ‘if’-clause is positioned within the underlying verb phrase, as reflected in (74).

(72) Watching movies if school is cancelled (for more than an hour) is forbidden.

(73) Imagine (me) ringing the bell if there’s a fire (very loudly).

(74)



Such nominalizations support the conclusion from §2.1 that ‘if’-clauses can combine with a verbal predicate at LF.

However, not all nouns embed verbal structure. Thus far we have ignored the distinction between roots √ (N/V) and elements v/n that mark nominal/verbal environments.[25] In the North Sámi example in (75), the elements distinguishing nominal and verbal 'work' are overt.[26] The noun isn't derived from a verbal form; the noun and verb are each derived from a common lexical core √BARG, and a nominalizing or verbalizing element, as reflected in (76).

(75) a. *barg**u*** ('work' (n.))
b. *barg**at*** ('work' (v.))

(76) a.
nP
n √BARG
/u/

b.
vP
v √BARG
/ɑ/

(cf. Julien 2015: 6; North Sámi (Uralic))

Like in English, a root-derived verb may itself be nominalized, as in (77b)–(77c) with Finnish √HAAST 'challenge'.[27]

(77) a. haast-**e**
√challenge-n^0
'a/the challenge'

b. haast-**a**-a
√challenge-v^0-INF
'to challenge'

c. Opiskelijoiden haast-**a-minen** on vaikeaa.
students.GEN √challenge-v^0-n^0 is difficult.PART
'Challenging students is difficult.' (Finnish)

However, the nominals used in *N if S* conditionals needn't have verbalizing morphology. The *N if S* conditional in (78) uses the root-derived noun form *haaste* 'challenge' from (77a).[28]

[25] See n. 2; see Borer 2013 for an alternative approach to category determination of roots.

[26] The morphemic order is derived by head movement. The final *-t* in (75b) is the infinitival marker.

[27] Thanks to Jussi Suikkanen for assistance with the data.

[28] The nominalizing /e/ vowel is doubled before possessive suffixes for phonological reasons. I ignore more fine-grained morphemic analyses of the other predicates.

(78) [Haast-ee-mme jos lämpötila laskee] on olla jäätymättä
challenge-n^0-1PL.POSS if temperature drops is be freeze.INFIII.ABE
hengiltä.
lives.ABL
'Our challenge if the temperature drops will be not to freeze to death.'
(Finnish)

N if S conditionals cannot be reduced to nominalizations of conditionalized verbal predicates.

The noun in (78) wears its root+nominalizer structure on its sleeve. For languages such as English, there are other tests for the presence/absence verbal categories (see also Arad 2005, Borer 2013, 2014; see Alexiadou & Borer 2020 for an overview). For instance, it has been argued that only verbs or verb-derived nominals accept (a)telicity modifiers, such as 'in under an hour' in (79)–(80) (adapted from Borer 2014: 72). Whereas the "argument structure" nominalization with 'perform*ance*' and ACC-*ing* nominalization with 'perform*ing*' in (80b)–(80c) can be modified by 'in under an hour', (80a) with 'recital' is anomalous.

(79) Alice performed the recital/concerto in under an hour.

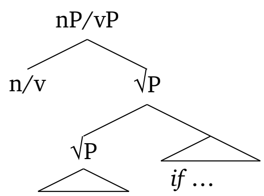
(80) a. *Alice's recital in under an hour (surprised us.)
b. Alice's performance of the recital/concerto in under an hour (surprised us.)
c. Alice performing the recital/concerto in under an hour (surprised us.)

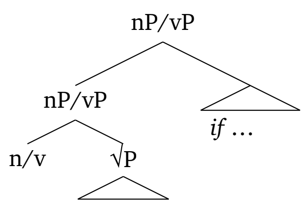
These contrasts have been taken as evidence that nouns such as 'recital' are derived directly from the root. In (80a), the thought goes, the verbal categories that delineate the complex event structure to be modified by 'in under an hour' are absent. An example of an *N if S* conditional with (root-derived) 'recital' is in (81).

(81) Your recital if you win the competition will be at Carnegie Hall.

I conclude that 'if'-clauses can modify verb phrases as well as (eventive and non-eventive) noun phrases.

This conclusion is insufficiently precise in the current framework. Both LFs in (82) are compatible with the data thus far. The analysis in (82a) treats conditional predicates as realizations of a common √P+'if'-clause structure in alternative nominal and verbal environments. The analysis in (82b) treats 'if'-clauses as possibly modifying both noun phrases and verb phrases.

(82) a. 

b. 

The following sections examine evidence from English and other languages for the alternative LFs in (82).

3.2 Conditionalized verbal predicates

Languages such as Persian and British English have been argued to allow a kind of verb phrase ellipsis in which v is pronounced ("v-stranding ellipsis"). Start with Persian (§2.1.3). Persian features a productive system of complex predicates formed from a light verb and a nonverbal predicate ("light verb constructions"). Examples with *kardan* 'to do' and *zadan* 'to hit' are in (83). The light verb has been argued to realize v (Folli et al. 2005, Karimi 2005). (I use small caps and subscript 'v' in glosses of light verbs to distinguish them from their "heavy" counterparts.)

(83) a. chune zadan
chin HIT_{v}
'to negotiate'

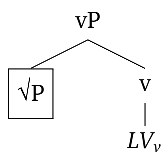
b. tamiz kardan
clean DO_{v}
'to clean' (adapted from Karimi 1997: 278–279; Persian)

Toosarvandani (2009, 2019) observes that the light verb can be pronounced in verb phrase ellipsis. In the second conjunct of (84), the nonverbal predicate *piranâro otu* 'iron the shirts' is elided but the light verb *zad* 'HIT_{v}' is pronounced. (Persian is verb-final with a basic SOV word order.)

(84) Sohrâb piranâ-ro otu na-zad, vali Rostam ~~[piranâ-ro otu]~~ **zad**.
Sohrab shirt.PL-OBJ iron NEG-HIT$_{V}$.PAST.3SG but Rostam shirt.PL-OBJ iron HIT$_{V}$.PAST.3SG
'Sohrab didn't iron the shirts, but Rostam did.'
(adapted from Toosarvandani 2009: 65; Persian)

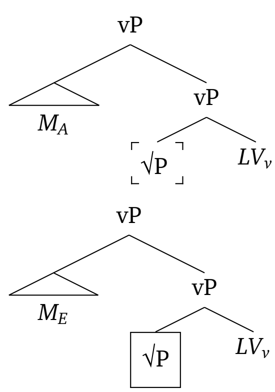
Toosarvandani argues that the ellipsis site is the complement of the light verb v — in the present framework, the √P, as indicated in (85) (using 'LV_v' for the light verb).[29]

(85)



Suppose this analysis of Persian v-stranding ellipsis is correct. We can use the construction to test the position of an 'if'-clause in a *V if S* conditional. Schematically, let M_A, M_E be different 'if'-clauses in the clause containing the antecedent and the clause containing the ellipsis, respectively. If the 'if'-clause modifies vP (or higher), then it should be possible to strand M_E with the light verb, using M_E LV_v. The modifier isn't in the ellipsis site, as reflected in (86).

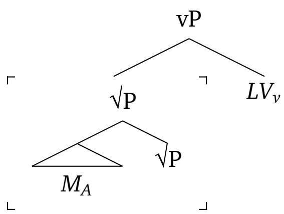
(86) a. b.



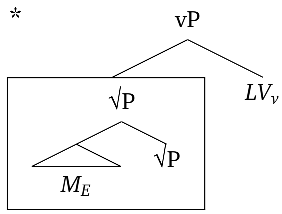
[29]Harley (2017) argues that Persian is a "Voice-bundling" language (Pylkkänen 2008), where the verbalizing function of v and external-argument-introducing function of v* (in my notation) are combined in a single head. I ignore this complication; vP may be replaced by v′ where appropriate. (On the need to distinguish v and v* in English, see Borer 2003, Harley 2009.)

If the ‘if’-clause modifies √P, however, stranding M_E with the light verb should be degraded. The representations of the ellipsis site and its antecedent in (87) are non-identical, and the ‘if’-clause M_E is incorrectly included in the ellipsis.

(87) a.

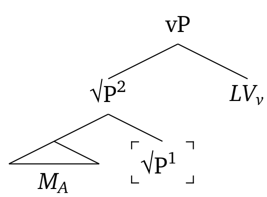


b. *



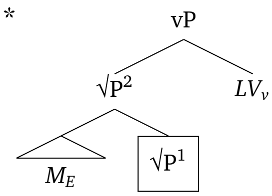
Removing the lower √P segment from the ellipsis site, as in (88), would satisfy the identity condition on ellipsis, though it doesn’t yield a possible target for ellipsis. It is commonly accepted that ellipsis targets the complement of a head that stands in a suitable (possibly trivial) relation to an ellipsis licensor (Lobeck 1995, Merchant 2001, 2004, Aelbrecht 2010). Yet $\surd P^1$ in (88b) isn’t the complement of v — or of any other head — to mark it for ellipsis.[30]

(88) a.



[30] See Merchant 2013: 102–103 for a parallel argument to explain the absence of the restitutive reading of ‘again’ in English verb phrase ellipsis. Surprisingly, Toosarvandani (2009: 77) assumes a structure isomorphic to Merchant’s (and to (87b)) for the restitutive reading but in an account of how the reading *is* possible in Persian v-stranding ellipsis. It isn’t said how the analysis is compatible with Toosarvandani’s feature-based approach and claim that “it is the complement of v … that is deleted” (2009: 61). One way around the argument would be if M_E could be inserted countercyclically (late merged with a node that isn’t the root) after $\surd P^1$ is marked for ellipsis by v. (See Lebeaux 1988, Chomsky 1993, Fox & Nissenbaum 1999 on possible countercyclic merger of adjuncts; for critical discussion see Sportiche 2019.) Working out the details would require taking a stand on controversial issues regarding the timing of ellipsis and its implementation in the syntax and PF interface. I will put it aside and assume that structures such as (88b) don’t represent possible targets for ellipsis.

b. *



The sentence in which the ‘if’-clause M_E is stranded with the light verb won’t be generated.

What we find, according to my consultants, is that stranding the ‘if’-clause with the light verb is possible in (89)–(90).[31]

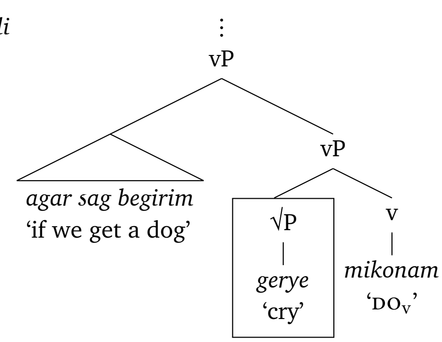
(89) Man agar gorbe begirim gerye ne-mikonam, vali agar sag
I if cat take.PRES.1PL crying NEG-DO$_V$.PRES.1SG but if dog
begirim ~~gerye~~ mikonam.
take.PRES.1PL crying DO$_V$.PRES.1SG
‘I won’t cry if we get a cat, but I will if we get a dog.’

(90) *A:* To agar piâz dârim âshpazi mikoni?
you if onion have.PRES.1PL cooking DO$_V$.PRES.2SG
‘Will you cook if we have onions?’

B: Na, vali agar sir dârim ~~âshpazi~~ mikonam.
no but if garlic have.PRES.1PL cooking DO$_V$.PRES.1SG
‘No, but I will if we have garlic.’ (Persian)

These data provide evidence that the ‘if’-clause is above √P. A representation of the second conjunct of (89) is in (91). The ‘if’-clause isn’t in the ellipsis site and can be pronounced with the light verb.

(91) *. . . vali*



[31] Thanks to Reza Hadisi, Mohsen Moghri, and Nader Shoaibi for assistance with the data.

Treating the 'if'-clause as combining with vP coheres with the Persian data from §2.1.3 indicating that the 'if'-clause is below an agentive subject.

British English (BE) has also been argued to have v-stranding verb phrase ellipsis — the "British *do*" construction, illustrated in (92) (Baker 1984, Thoms 2019: 1032–1038).[32] Following a finite auxiliary with a nonfinite form of 'do' in this way is strongly ungrammatical in American English.

(92) a. Alice will bring snacks, and Bert will **do**, too.
b. I'm not sure if Alice brought snacks, but she may have **done**. (BE)

It has been argued that British 'do' also realizes v and elides its complement in such sentences (Aelbrecht 2010, Thoms 2011, Baltin 2012).[33] For instance, (92b) suggests that 'do' is under aspectual auxiliaries. From the other direction, examples such as (93) where 'do' co-occurs with an unaccusative configuration provide evidence that 'do' is above √P (cf. Baltin 2012: 388). The internal argument ('Bert') is extracted from the ellipsis site into subject position, as reflected in (93b) (cf. §2.1.2). British 'do' is above the (elided) √P in which the pronounced subject originates.

(93) a. Alice might fall, and Bert might do, too. (BE)
b. Alice$_1$ might [fall t_1], and Bert$_2$ might do ~~[fall t_2]~~, too.

'If'-clauses can also survive ellipsis with British 'do', as in (94)–(95).[34]

[32]Thanks to Amelia Combes, Alice Corr, Amit Ginbar, Terry Goetz, Anya Saliy, Sascha Sturgeon, and Gary Thoms for discussion and assistance with the data.

[33]For critical discussion see also Thoms 2019, Lewis 2022, Silk 2023. That the 'do' in this construction isn't a main verb (or root) can also be seen from examples such as (i) (cf. Stroik 2001: 364–365). As the antecedent of the *wh* object of 'do' is a full predicate ('buy candy'), the phrase headed by 'do' must be larger than such a predicate, i.e. larger than √P.

(i) Bert might buy candy, which he shouldn't do.

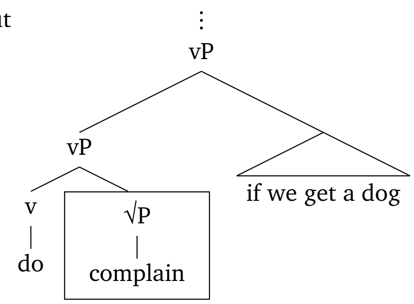
[34]The informal judgments from my consultants were confirmed in an acceptability judgment task. Participants were 32 native British English speakers (recruited from Prolific and the University of Birmingham). The experiment had a Latin square design with factors for the type of ellipsis (VPE vs. British 'do') and the presence/absence of a stranded 'if'-clause. 8 lexicalizations (tokens) of each sentence type (condition) were distributed among 4 lists via a Latin square procedure, so each participant saw two items for each condition. 4 orders for each list were created and pseudorandomized so that experimental items didn't appear sequentially, yielding 16 surveys in total (2 participants per survey). Each survey included the same practice sentences, filler sentences, and attention checks. The sentences were rated from 1–7. Each participant's ratings (excluding practice items) were z-score transformed to reduce scale bias. No evidence was found of a violation from stranding an 'if'-clause with British 'do' ellipsis. The interaction between 'if'-clause stranding and British 'do' was not significant ($F(1, 27) = 0.08$, $p = 0.78$). The mean z-score rating for sentences

(94) Leslie will complain if we get a cat, but Harper may do if we get a dog.

(95) George should milk the cows if it's a weekday, but Tracey should do if it's a Saturday. (BE)

As previously, positing that the 'if'-clause combines with vP provides a natural explanation. Like in (91), the 'if'-clause in (96) isn't part of the ellipsis site.

(96) … but



3.3 Conditionalized nominal predicates

Layered analyses of noun phrases have been less well developed in comparison to layered analyses of verb phrases. Differences in aims and notation can make it difficult to translate theories of noun phrase ellipsis, anaphora, and modification into the present architecture. Conclusions drawn in what follows about the position(s) of 'if'-clauses on the basis of such phenomena must be provisional.

§3.2 examined v-stranding ellipsis data in which v is pronounced. Maeda & Takahashi (2016, 2017) have argued that the analogous case of n-stranding ellipsis, where an overt nominalizing element n survives noun phrase ellipsis, is found in dialects of Japanese. Examples from the Nagasaki dialect are in (97)–(98). The light noun *to* (≈ 'one'), Maeda & Takahashi argue, realizes n and elides its complement.[35] In (98) the internal argument 'flute' of the elided noun 'handling' is extracted from

like (94)–(95) with a stranded 'if'-clause and British 'do' was positive; the mean non-transformed rating was 5.8 (SEM = .15).

[35] Maeda & Takahashi (2017) extend the analysis to Oita and Toyama dialects. See Saab 2019: 557–559 on possible extensions to certain Dutch dialects. I leave open whether the analysis is appropriate for English 'one' (see Llombart-Huesca 2002, Harley 2005). For neutrality I gloss *n(o)* as a linker (LNKR); its analysis is controversial (e.g., Saito et al. 2008). Like with British 'do' ((ii)) — and unlike with Persian light verbs ((iii)) — ellipsis is obligatory with Nagasaki Japanese *to* ((i)):

(i) Haruna-n taido-wa Mariko-n (*taido) to (*taido) yorimo rippayatta.
Haruna-LNKR attitude-TOP Mariko-LNKR attitude ONE$_{N}$ attitude than good
(adapted from Maeda & Takahashi 2016: ex. 22; Nagasaki Japanese)

(ii) Sohrab won't iron the shirts, but Rostam will do (*iron the shirts). (BE) the ellipsis site (cf. (93)). (Japanese is head-final with a basic SOV word order. I gloss *to* with ‘ONE$_N$’ to distinguish it from English ‘one’.)

(97) Haruna-n taido-wa Mariko-n ~~taido~~ **to** yorimo rippayatta.
Haruna-LNKR attitude-TOP Mariko-LNKR attitude ONE$_N$ than good
lit. ‘Haruna’s attitude was better than Mariko’s one.’

(98) Haruna-n piano-n toriatukai-wa teineiya kedo, Mariko-n
Haruna-LNKR piano-LNKR handling-TOP careful though Mariko-LNKR
furuuto-n$_1$ ~~[t_1 toriatukai]~~ **to**-wa sozatuya ne.
flute-LNKR handling ONE$_N$-TOP rough PART
lit. ‘Though Haruna’s handling of the piano is careful, Mariko’s one of the flute is rough.’
(adapted from Maeda & Takahashi 2016: exs. 7, 14–15; Nagasaki Japanese)

My consultants report that stranding an ‘if’-clause with *to* is possible under nominal ellipsis.[36] Examples with the conditional marker *baai* are in (99)–(100).

(99) [Haruna-no kanozyo-no chiimu-ga katu baai-no paatii]-wa [Asa-no
Haruna-LNKR her-LNKR team-NOM win case-LNKR party-TOP Asa-LNKR
kanozyo-no chiimu-ga katu baai-n ~~paatii-n~~ to] yori tanosii.
her-LNKR team-NOM win case-LNKR party-GEN ONE$_N$ than fun
‘Haruna’s party if her team wins will be more fun than Asa’s if her (Asa’s) team wins.’

(100) [Dairinin-ga kuru baai-no tetuduki]-wa [honnin-ga kuru
agent-NOM come case-LNKR process-TOP the.person-NOM come
baai-n ~~tetuduki-n~~ to] yori hukuzatu.
case-LNKR process-GEN ONE$_N$ than complicated
‘The process if an agent comes is more complicated than the one if the person comes.’
(Nagasaki Japanese)

(iii) Sohrab piranâ-ro otu ne-mizane, vali Rostam (piranâ-ro otu) mizane.
Sohrab shirt.PL-OBJ iron NEG-HIT.V.PRES.3SG but Rostam shirt.PL-OBJ iron HIT.V.PRES.3SG
‘Sohrab won’t iron the shirts, but Rostam will (iron the shirts).’ (Persian)

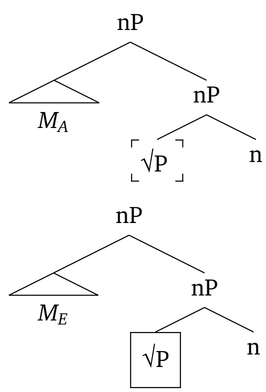
More detailed comparisons would be worth exploring.

[36] Thanks to Masako Maeda for assistance with the data.

In (100), ‘process’ is elided and the conditional clause ‘if the person comes’ is stranded with the light noun *to*.

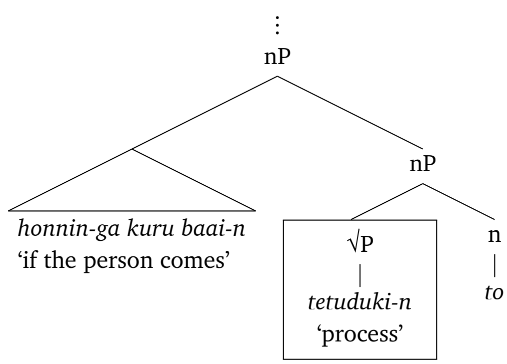
Suppose Maeda & Takahashi are correct that Nagasaki Japanese *to* realizes n and elides its complement. The line of argument from §3.2 for the case of ‘if’-clauses and v-stranding ellipsis carries over. The acceptability of (99)–(100) is expected if the ‘if’-clause combines above √P. The ‘if’-clauses M_A, M_E in (101) aren’t part of the material that needs to be identical in order for the ellipsis to be licensed (cf. (86)).

(101) a. b.



An analysis of the noun phrase containing the ellipsis from (100) is in (102). The predicate *tetuduki* ‘process’ is elided under identity with its antecedent, stranding the n head *to* and higher conditional *baai*-clause.

(102)



Treating the ‘if’-clause as combining with nP coheres with the arguments from §2.2.4 that the ‘if’-clause is under a thematic possessor.

3.4 Conditionalized lexical predicates

The previous sections provided evidence from ellipsis that ‘if’-clauses can be interpreted as vP or nP modifiers, per (82a). This section examines evidence for the alternative structure in (82b) where the ‘if’-clause combines under v or n. I begin with data from modifiers that have been argued to be interpreted low at the level of the √P.

First, it has been observed that sentences such as (103) with ‘again’ can have multiple interpretations — notably, one implying the repetition of an event of closing the door, brought out in (103a) (also called the “repetitive reading”), and another interpretation implying the repetition of the state of the door being closed, brought out in (103b) (also called the “restitutive reading”).

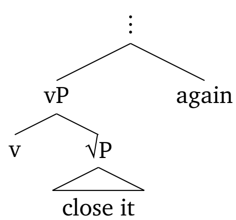
(103) Bert closed the door again.

a. Alice closed the door. It blew open. Bert closed it again. (*event reading*)

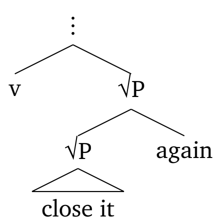
b. The door was closed. The wind blew it open and no one closed it. Finally, Bert closed it again. (*state reading*)

It is common to analyze the different readings as reflecting a syntactic ambiguity (von Stechow 1996, Beck & Johnson 2004, Johnson 2008; see also Beavers & Koontz-Garboden 2020, Yu et al. 2023). Alternative LFs in the present framework are in (104a). Roughly speaking, the state reading (103b) arises when ‘again’ combines low under v and modifies the √P, which describes a state of the door being closed; and the event reading (103a) arises when ‘again’ combines above v and modifies the vP, which describes an event of resulting in such a state.

(104) a. (*event reading*)



b. (*state reading*)

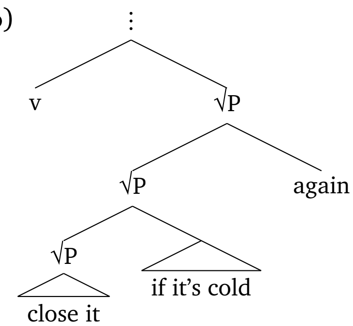


In (105) we see ‘again’ modify a *V if S* conditional. The relevant reading is the state reading which implies a previous state of the window being closed if it was cold outside.[37] (105) could be used in a context where the window is new and one doesn't know whether it has ever been closed before.

(105) Last night, the window was closed if it was cold outside. Today I saw it open and no one has closed it. Tonight, I will close it if it's cold out again.

An LF as per the analysis in (104b) is in (106).

(106)



The 'if'-clause combines low with √P under v.

A similar pattern can be observed with temporal 'for' modifiers such as 'for five minutes' in (107). On the event (or "durative") reading, brought out in (107a), 'for five minutes' characterizes the event of closing the window; on the state (or "internal") reading, brought out in (107a), 'for five minutes' characterizes the state of the window being closed.

(107) Bert closed the window for five minutes.

---

[37] The English word order may make it difficult to exclude a reading where 'again' is interpreted inside the 'if'-clause. In the Persian examples in (i)–(ii), *dobâre* 'again' is clearly outside the 'if'-clause.

(i) Dishab agar birun sard bude panjere baste bude. Emruz didam un bâz-e va
last night if outside cold was window closed was today saw.1SG it open-is and
hichkas na-baste-esh. Emshab dobâre [agar sard bashe] mibandam-esh.
no one NEG-closed-it tonight again if cold be.SUBJ.3SG close.PRES.1SG-it
=(105)

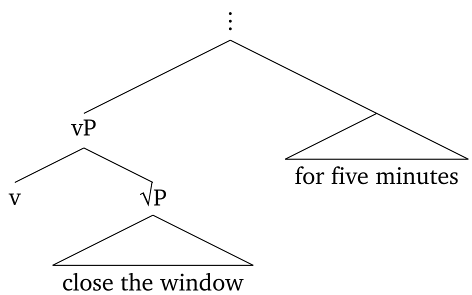
(ii) Dishab agar mehmunha dâshtan miumadan âshpazxune tamiz bude. Alân
last night if guests had.3PL came.IMPFV.3PL kitchen clean was now
âshpazxune kasif-e. Emshab dobâre [agar mehmunha miyân] tamiz-esh mikonam.
kitchen dirty-is tonight again if guests come.PRES.3PL clean-it do.PRES.1SG
'Last night, the kitchen was clean if guests were coming over. Now the kitchen is dirty. Tonight, I will clean it [if guests are coming over] again.'

a. *Event reading* ≈ Bert spent five minutes closing the window. (He was moving really slowly.)

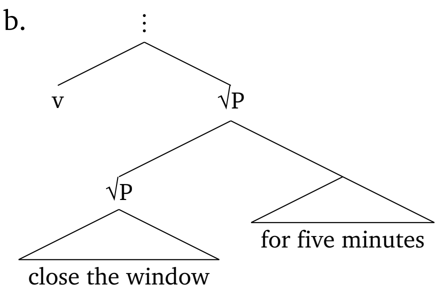
b. *State reading* ≈ Bert closed the window, and it stayed closed for five minutes.

The different readings can again be represented structurally. In (108b) 'for five minutes' is interpreted low and modifies the √P; the state described by the √P of the window being closed is measured as lasting five minutes. In (108a) the modifier is interpreted above v and measures the event of resulting in a state of the window being closed.

(108) a. (*event reading*)



b. (*state reading*)



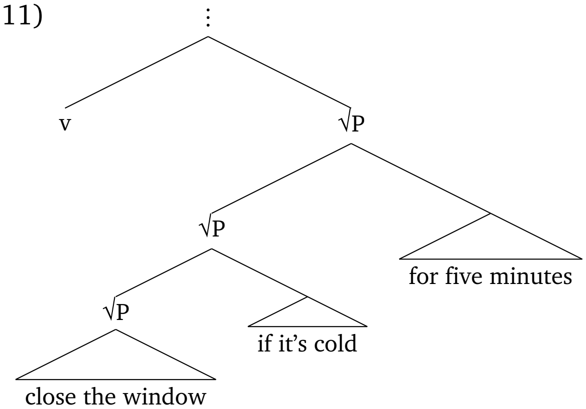
In (109)–(110) the temporal 'for' phrase modifies the *V if S* conditional. The relevant reading is, again, the state reading, where the 'for' phrase characterizes the state described by the √P. In (110) the 'for' phrase measures the duration of the defendants' hypothetical imprisonment.

(109) Bert will close the window if it's cold outside for five minutes.

(110) The judge will imprison the defendants if they're found guilty for the maximum sentence.

An analysis for (109), following (108b), is in (111). The 'for' phrase modifies the conditionalized √P predicate under v.

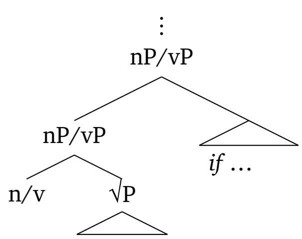
(111)



4 Taking stock

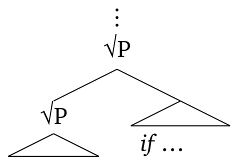
Let’s take stock. This paper has investigated a class of comparatively understudied conditional expressions: ‘if’-clauses in noun phrases and verb phrases (“conditional predicates”). I have focused on the structure of conditional predicates at LF. §2 provided evidence from ellipsis and interactions with modifiers that ‘if’-clauses can be interpreted under v* or n*, that is, under the base position of an external argument such as a thematic agent or possessor. §§3.2–3.3 provided evidence that the ‘if’-clause can be interpreted at the level of the verbal vP or nominal nP predicate. Crosslinguistic data from v-stranding ellipsis and n-stranding ellipsis show that ‘if’-clauses can survive ellipses which have been argued to target the complement of v and n. This supports that the ‘if’-clause is interpreted higher, as in (112).

(112)



§3.4 showed that ‘if’-clauses can be interpreted in the scope of √P modifiers such as state readings of ‘again’ and temporal ‘for’ phrases. This provides evidence that interpretation at the level of the √P is also possible, as in (113).

(113)



‘If’-clauses can combine with lexical predicates, noun phrases, and verb phrases. And they can modify predicates of events, states, and entities.

The data and syntactic results in this paper have rich implications for work on the semantics of conditionals, root meaning, and nominal and verbal predication. For instance, a longstanding debate in semantics is between “operator” theories that treat ‘if’ as a two-place operator, and “restrictor” theories that treat ‘if’ as marking a restriction on some external operator (Lewis 1973, Kratzer 2012). Extending such theories to *N if S* and *V if S* conditionals is nontrivial. One must ensure that the ‘if’-clause can combine at each level and modify predicates of diverse types (events, states, entities). Upstream, the result of combining the ‘if’-clause with its sister must eventually be able to combine with a sentential subject (e.g., ‘the students’) or a determiner (e.g., ‘the’). The challenges are not merely technical. The meaning of ‘if’ in conditional predicates should be plausibly related to its meaning in sentence-initial contexts. There are controversial issues regarding event structure, tense and aspect, and shifted interpretations of expressions outside the ‘if’-clause. I leave the investigation of such developments to future research.

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